MUNICIPAL CORPORATION OF GREATER MUMBAI (MCGM / BMC)

RED WARNING

TIER_2_SCREENING_SIMULATION

DISASTER MANAGEMENT CELL • EMERGENCY OPERATIONS EXECUTIVE DOSSIER

Document ID: **DOS-RAINFALL_WATERLOGGING-T2H-MVGNDTGT** • Generated: Tuesday, 20 October 2026 at 5:46 pm IST

EXECUTIVE SITUATION SUMMARY:

Multi-criteria hydrodynamic simulation evaluated for scenario **"Scenario A: Monsoon Downpour & Low-Lying Waterlogging"** across all 24 canonical administrative wards of Greater Mumbai over a **T+2h** forecast horizon. Topographical saucer basins and coastal backpressure were evaluated against Copernicus GLO-30 DEM terrain profiles.

SEVERE WARDS	HIGH RISK WARDS	CITYWIDE MEAN RISK	SCREENED POPULATION
1	11	54.9	~1588k
Score ≥ 80 / 100	Score 60 - 79 / 100	Normalised 0-100 scale	Low-lying contour buffers

1. KEY INTELLIGENCE FINDINGS

EVIDENCE-BASED SYNTHESIS

- Ward G/North (Dadar / Dharavi / Mahim) holds the highest hazard score of 84.4, classified as 'Severe Inundation Hazard', with an estimated exposed population of 180,000.
- The citywide total exposed population is estimated at 1,588,100, with a mean hazard score of 54.9 across 24 wards.
- Ward F/South (Parel / Sewri / Naigaon) and Ward F/North (Matunga / Sion / Wadala) both hold a score of 78.5, classified as 'High Sinking / Choke', with exposed populations of 70,200 and 95,400 respectively.
- The scenario parameters applied are 45 mm/h rainfall rate, 3-hour duration, and 15% drainage blockage, triggering a RED_WARNING alert level.

2. HIGHEST PRIORITY INTERVENTION WARDS

TOP 5 OF 24

RANK	WARD ID	WARD DESIGNATION	RISK SCORE	SEVERITY TIER	PRIMARY PHYSICAL TRIGGER
#1	Ward G/North	Ward G/North (Dadar / Dharavi / Mahim)	84.4 index (0-100)	SEVERE INUNDATION HAZARD	Low elevation + tidal backwater
#2	Ward F/South	Ward F/South (Parel / Sewri / Naigaon)	78.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#3	Ward F/North	Ward F/North (Matunga / Sion / Wadala)	78.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#4	Ward E	Ward E (Byculla / Nagpada / Mazgaon)	72.5 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#5	Ward L	Ward L (Kurla / Sakinaka / Asalpha)	69.7 index (0-100)	HIGH SINKING / CHOKE	Saucer depression ponding

3. POTENTIALLY EXPOSED MUNICIPAL INFRASTRUCTURE

CRITICAL ASSETS

Tertiary Municipal Hospitals — 7 assets

KEM Hospital, Sion Hospital, Nair Hospital, Cooper Hospital

Caveat: Located in spatial screening proximity (<500m); physical water ingress not confirmed.

Arterial Transit Corridors — 12 assets

Dr. Babasaheb Ambedkar Road, LBS Marg, Swami Vivekananda (SV) Road

Caveat: Road centerline intersects modeled basin buffer; traffic impassability subject to spot depth.

Copernicus GLO-30 Topographic Depressions **6 assets**

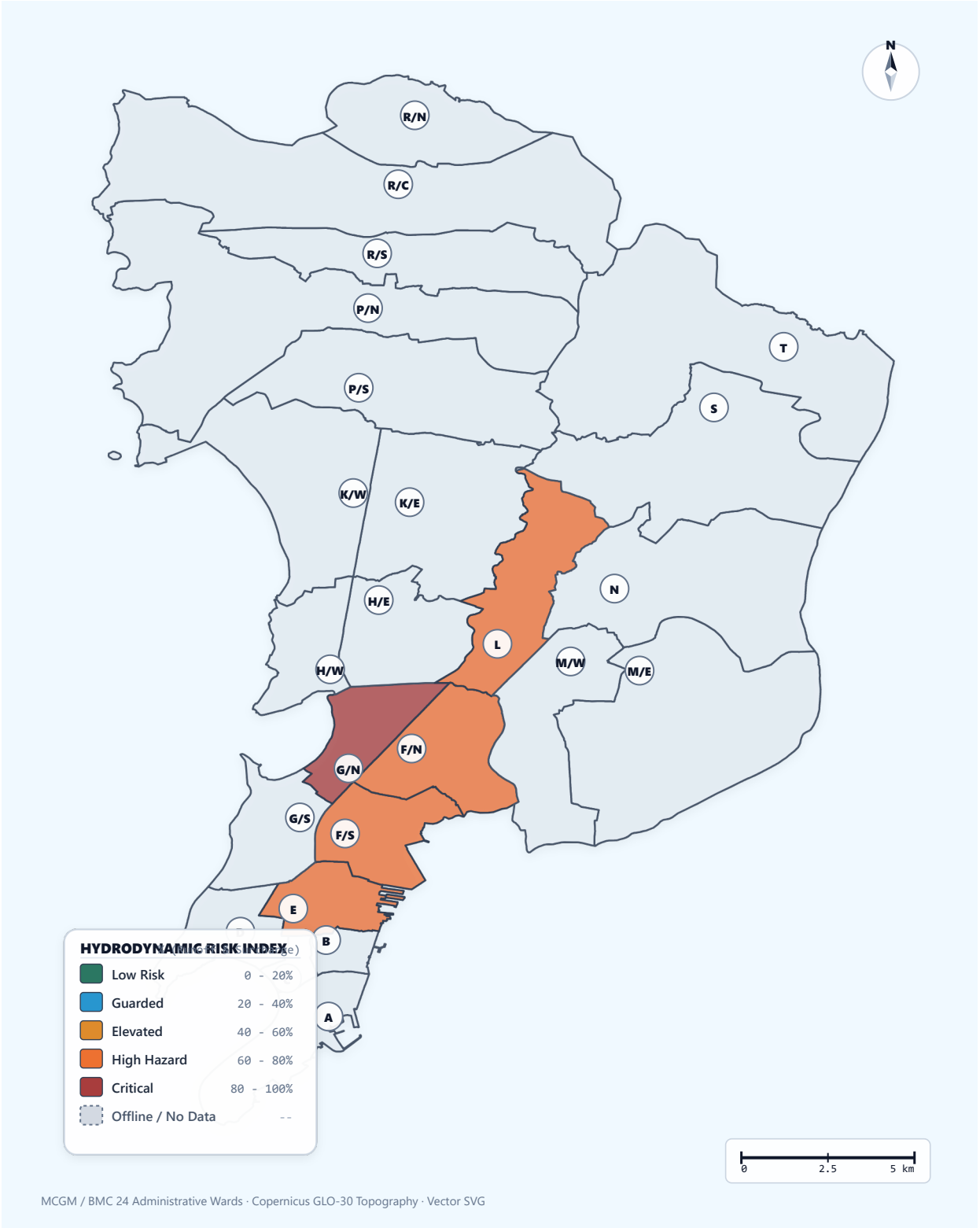
Hindmata Basin (4.5m ASL), Kurla Sump (5.2m ASL), Milan Subway (5.8m ASL)

Caveat: Documented elevation minima where gravity storm-water discharge reverses during high tide.

GREATER MUMBAI 24-WARD HYDRODYNAMIC CARTOGRAPHY

T+2h Simulation Horizon

FIGURE 1.1: MUNICIPAL CHOROPLETH RISK DISTRIBUTION & TOPOGRAPHIC SINK SCREENING



Cartographic Note: Wards are filled according to their calculated waterlogging and drainage surcharge score for the active scenario. Boundaries adhere to canonical BMC administrative boundaries (EPSG:4326). Blue boundary highlights selected ward focus. Shaded contours represent Copernicus GLO-30 DEM depression sinks.

COMPREHENSIVE 24-WARD MUNICIPAL EVALUATION MATRIX

24 BMC Administrative Wards

SECTION C: ADMINISTRATIVE WARD RANKINGS & POPULATION EXPOSURE

RANK	WARD	ADMINISTRATIVE WARD DESIGNATION	SCORE	SEVERITY	ELEV (M)	EXPOSED POP	ASSETS	PRIMARY TRIGGER
#1	G/North	Ward G/North (Dadar / Dharavi / Mahim)	84.4	SEVERE INUNDATION HAZARD	4.9m	~1,80,000	1	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#2	F/South	Ward F/South (Parel / Sewri / Naigaon)	78.5	HIGH SINKING / CHOKE	4.8m	~70,200	6	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#3	F/North	Ward F/North (Matunga / Sion / Wadala)	78.5	HIGH SINKING / CHOKE	5.1m	~95,400	3	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#4	E	Ward E (Byculla / Nagpada / Mazgaon)	72.5	HIGH SINKING / CHOKE	6.2m	~79,200	4	Score = [RunoffExcess(20.1 mm/h over Cap 21.3) → 1
#5	L	Ward L (Kurla / Sakinaka / Asalpha)	69.7	HIGH SINKING / CHOKE	6.9m	~1,62,000	5	Score = [RunoffExcess(0.0 mm/h over Cap 42.5) → 0.

* Population estimates derive from 2011 Census ward totals adjusted to 2026 BMC projections, intersected with low-lying saucer depression buffer zones.

MATHEMATICAL METHODOLOGY & INPUT CONDITIONS

SECTION E: FORMULA SPECIFICATION, TERRAIN INTEGRATION & AUDIT TRACE

1. MATHEMATICAL FORMULATION

HYDRODYNAMIC SCREENING

```
S_pond(t) = min(100, 0.35 * (P_excess(t) / 45 * 100) + 0.25 * D_topo + 0.20 * H_hotspot + 0.15 * d_drainage + 0.05 * S_accum(t))
```

Latex Representation: `S_pond(t) = min(100, 0.35 * (P_excess(t) / 45 * 100) + 0.25 * D_topo + 0.20 * H_hotspot + 0.15 * d_drainage + 0.05 * S_accum(t))`
Baseline Definition: Dry-weather baseline assuming 0 mm/h rain, 1.5m astronomical tide, 0% drainage blockage, and historical hotspot background. Delta percent represents: ((Score - Baseline) / max(1, Baseline)) * 100%.
Spatial Resolution: 24 Canonical BMC Administrative Wards + 250m Continuous Raster Grid • **Temporal Resolution:** Hourly increments from T+0h to T+6h

2. USER-CONTROLLED SCENARIO INPUTS & OBSERVATIONS

PARAMETER AUDIT

INPUT PARAMETER / VARIABLE	VALUE	UNIT	STATUS	SOURCE / NOTES
Scenario Precipitation Rate	45	mm/h	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Event Duration	3	hours	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Culvert & Nullah Siltation	15	%	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Temporal Rainfall Profile	0	profile	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Automatic Weather Station (AWS) Precipitation (Live Telemetry)	0	mm/h	ACTIVE	Open-Meteo High-Resolution AWS Feed
Apollo Bandar Astronomical Tide Level (Live Telemetry)	1.85	meters Chart Datum (CD)	ACTIVE	Mumbai Port Trust (MbPT) Tide Gauge Table

3. MISSING OR OFFLINE TELEMETRY FEEDS

HONEST DISCLOSURE

VARIABLE NAME	IMPACT ON SIMULATION MODEL	DETERMINISTIC FALLBACK USED
Real-time Drainage Sluice Gate SCADA	Outfall gate blockage % is user-configured rather than live sensor-driven.	Scenario parameter default applied with explicit assumption note.
Mithi River Ultrasonic Level Sensors	River corridor ponding modeled via static Copernicus elevation buffer.	GLO-30 DEM 5.0m elevation contour threshold.

RECOMMENDED ACTION DIRECTIVES & STATUTORY DISCLAIMER

SECTION F: INTER-AGENCY OPERATIONS & EXECUTIVE SIGN-OFF

1. DEPARTMENTAL OPERATIONAL ACTION DIRECTIVES

PRIORITY INTERVENTIONS

PHASE	DEPT	ACTION TITLE & DIRECTIVES	TARGET WARDS	PRIORITY
BEFORE_EVENT	SWD	**Pre-emptive Pumping Station Readiness Audit** Conduct on-site inspection, functional testing and fuel reserve verification for all pumping stations and floodgates located within the screened zones of Wards G/North, F/South, F/North, E and L.	E, L	HIGH
DURING_MONITORING	SWD	**Real-time Water Level Monitoring at Depressions** Deploy portable water-level loggers at each named depression (e.g., Mahim Creek Mangrove Intertidal Basin, Hindmata-Parel Saucer Depression) and integrate data into the municipal command dashboard.	E, L	CRITICAL
POST_EVENT	SWD	**Post-event Culvert Desiltation and Inspection** Mobilize desiltation crews to clear silt buildup in culverts and nullahs identified in the scenario (15% blockage) across the affected wards.	E, L	HIGH
BEFORE_EVENT	DMD	**Update Land-Use and Zoning Maps for Chronic Hotspots** Incorporate the listed chronic hotspots into the official zoning database and flag them for future development restrictions or mitigation mandates.	E, L	MEDIUM
BEFORE_EVENT	ROADS	**Pre-position Sandbags and Traffic Diversion Plans for Key Arterials** Stage sandbag kits and draft alternative routing maps for Senapati Bapat Marg, Dr. Babasaheb Ambedkar Road, LBS Marg, SCLR and AGLR segments intersecting the high-risk zones.	E, L	CRITICAL

2. AI GOVERNANCE & DATA PROVENANCE

AUDIT TRACE

AI Orchestration State: PARTIAL_DEGRADED (6997 ms)
Role 1 (Evidence Analyst): FAILED_AUTH (deterministic-rules-engine)
Role 2 (Impact Analyst): OPERATIONAL (openai/gpt-oss-120b)
Role 3 (Report Editor): OPERATIONAL (qwen/qwen3.8-27b)

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EXECUTIVE REVIEW & AUTHORISATION:

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Disaster Management Department, MCGM Head Office

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STATUTORY DISCLAIMER & LIMITATIONS: This document constitutes a computerized multi-criteria hydrodynamic screening evaluation based on digital elevation modeling (Copernicus GLO-30 DEM) and BRIMSTOWAD storm drainage catchments. It is designed to assist administrative situational awareness and pre-positioning. It does NOT constitute an automated municipal proclamation, legal evacuation order, or public emergency dispatch. Official deployment of personnel, equipment, and public alarms requires formal sign-off by the Municipal Commissioner or designated Disaster Management Officer.